INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

Infosys SE Complete Preparation — Time, Speed, Distance, Trains & Boats

Section: prepinsta infosys SE | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

Infosys SE Complete Preparation — Time, Speed, Distance, Trains & Boats

Source Portal: PrepInsta Infosys Preparation Suite (`https://prepinsta.com/infosys/aptitude/speed-time-and-distance/`)
Target Role: Infosys Systems Engineer (SE) | **Category:** 01_Mathematical_Ability_Technical_Aptitude
Total Verified Questions Extracted: 11
Quizzes Covered: quiz-1, quiz-2, quiz-3, quiz-4, quiz-5, quiz-6, quiz-7, quiz-8

📋 Section 1: Rules to Keep in Mind

These rules and fundamental principles govern all Infosys questions on this topic:

- Speed = Distance / Time; Distance = Speed * Time; Time = Distance / Speed.
- Conversion: 1 km/hr = 5/18 m/s; 1 m/s = 18/5 km/hr.
- Average Speed for two equal distances = 2xy / (x + y), where x and y are speeds.
- Relative Speed: In opposite directions = S1 + S2; In same direction = |S1 - S2|.
- Train crossing a pole/person: Distance = Length of train.
- Train crossing a platform/bridge: Distance = Length of train + Length of platform.

⚡ Section 2: How to Solve Quickly

Proven time-saving strategies recommended by PrepInsta for Infosys 2026 examination:

1. **Option Elimination via Unit Digits & Multiples:** Before doing full calculations, inspect the options to see if last digits or divisibility by 3, 7, 9, 11 can eliminate 2 to 3 wrong choices.
2. **Approximation & Magnitude Checks:** When exact arithmetic is tedious, round numbers to neat multiples (e.g., 99 ≈ 100) to find the nearest valid option.
3. **Reverse Option Substitution:** In complex algebraic equations, plug option values directly into the question stem to test validity.

🚀 Section 3: Tricks & Shortcuts

Key mathematical formulas, mental-math shortcuts, and speed axioms:

- Downstream speed (with current) = u + v; Upstream speed (against current) = u - v.
- Speed of boat in still water = (Downstream + Upstream) / 2.
- Speed of stream / current = (Downstream - Upstream) / 2.

📚 Section 4: Practice Questions, Options & Step-by-Step Solutions

Contains **11** verbatim questions with options, verified answer keys, and complete derivations:

Question 1 (PrepInsta quiz-1 - Q1)

Question Stem:
A motorboat, whose speed in 15 km/hr in still water goes 30 km downstream and comes back in a total of 4 hours 30 minutes. The speed of the stream (in km/hr) is

Options:
- **(A)** `2 km/hr`
- **(B)** `3 km/hr`
- **(C)** `4 km/hr`
- **(D)** `5 km/hr`

Correct Answer: (D) `5 km/hr`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Let the speed of stream be x km/hr,

then,

30/(15-x) +30/(15+x) = 4.5

450-30x +450+30x = 4.5(225-x2)

900/4.5= 225 - x2

200-225 = -x2

X = 5km /hr
```

Question 2 (PrepInsta quiz-1 - Q2)

Question Stem:
If Rahul rows 15 km upstream in 3 hours and 21 km downstream in 3 hours, then the speed of the stream is

Options:
- **(A)** `5 km/hr`

- **(B)** `4 km/hr`
- **(C)** `2 km/hr`
- **(D)** `1 km/hr`

Correct Answer: (D) `1 km/hr`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Rate upstream = (15/3) kmph


Rate downstream (21/3) kmph = 7 kmph.


Speed of stream (1/2)(7 - 5)kmph = 1 kmph
```

Question 3 (PrepInsta quiz-1 - Q3)

Question Stem:

A man rows 750 m in 675 seconds against the stream and returns in 7 and half minutes. His rowing speed in still water is

Options:

- **(A)** `4 kmph`
- **(B)** `5 kmph`
- **(C)** `6 kmph`
- **(D)** `7 kmph`

Correct Answer: (B) `5 kmph`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Rate upstream = (750/675) = 10/9 m/sec


Rate downstream (750/450) m/sec = 5/3 m/sec


Rate in still water = (1/2)*[(10/9) + (5/3)] m/sec.


= 25/18 m/sec


= (25/18)*(18/5) kmph


= 5 kmph
```

Question 4 (PrepInsta quiz-1 - Q4)

Question Stem:
A boat can travel with a speed of 16 km/hr in still water. If the rate of the stream is 5 km/hr, then find the time taken by the boat to cover the distance of 84 km downstream.

Options:
- **(A)** `4 hours`
- **(B)** `5 hours`
- **(C)** `6 hours`
- **(D)** `7 hours`

Correct Answer: (A) `4 hours`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
It is very important to check, if the boat speed given is in still water


or with water or against water. Because if we neglect it


we will not reach on right answer. I just mentioned here because


mostly mistakes in this chapter are of this kind only.


Lets see the question now.


Speed downstream = (16 + 5) = 21 kmph


Time = distance/speed = 84/21 = 4 hours
```

Question 5 (PrepInsta quiz-1 - Q5)

Question Stem:
One gear pulley rotates at a speed of 3 rev/sec another gear roates at 5 rev/sec. if both start together after how many seconds will they be together again?

Options:
- **(A)** `3 sec`
- **(B)** `20 sec`
- **(C)** `15 sec`
- **(D)** `5 sec`

Correct Answer: (C) `15 sec`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
3 rotates 3,6,9,12,15

5 rotates 5,10,15

because ans is 15
```

Question 6 (PrepInsta quiz-1 - Q6)

Question Stem:
A train leaves Meerut at 5 a.m. and reaches Delhi at 9 a.m. Another train leaves Delhi at 7 a.m. and reaches Meerut at 10.30 a.m. At what time do the two trains travel in order to cross each other ?

Options:
- **(A)** `08:30`
- **(B)** `07:25`
- **(C)** `07:56`
- **(D)** `none`

Correct Answer: (C) `07:56`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
First train takes 4 hours and the second train takes 3.5 hours.

Time ratio is 8:7. Therefore, the speed ratio will be 7:8.

Let the speeds be 7x and 8x, and distance be 28x ( 4x7 or 3.5x8).

At 7 AM, the first train must have covered a distance of 14x.

Therefore, at 7 A.M. the distance between the two trains is 28x-14x=14x.

Time taken to meet = 14x/(7x+8x)=14/15 hour or 56 minutes.

Hence, the two trains meet at 7.56 AM.
```

Question 7 (PrepInsta quiz-1 - Q7)

Question Stem:
An express A starts at 2.30 pm from nampali station and travels towards vizag station at speed of 80 kmph. Another expressstarts at 4.30 pmfrom nampali station to vizag station at speed of 100 kmph. How far awayfrom station Nampali will the two trains meet?

Options:

- **(A)** `A.600 km`
- **(B)** `B.700 km`
- **(C)** `C.750 km`
- **(D)** `D. 800 km`

Correct Answer: (D) `D. 800 km`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Let train A take x hours to reach the point of meeting(let it be P)

Then train B will reach the same point in

x-2 hours.

Distance covered to reach P by train A in x hours=Distance covered by B to reach P in x-2 hrs.

80*x=100(x-2)

80x=100x-200

x=10 hrs.

Distance from Nampali=x*speed of train A

=10*80

=800 km
```

Question 8 (PrepInsta quiz-1 - Q8)

Question Stem:

A journey of 600 km, Due to some problem in Vehicle speed was reduced to 200 kmph and it takes 30min extra, Find the Actual time taken for Journey?

Options:

- **(A)** `1 hour`
- **(B)** `1 hour 10 mins`
- **(C)** `2 hour 40 mins`
- **(D)** `45 mins`

Correct Answer: (A) `1 hour`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Here the Distance is constant. So, let the actual time is x.


Then,


600/x-600/(x+(1/2))=200


After we solve this we will get 1hr.


So, the actual time taken for the journey is 1hr.
```

Question 9 (PrepInsta quiz-1 - Q9)

Question Stem:
Two men start from opposite banks of a river . They meet 340 meters away from one of the banks on forward journey. After that they meet at 170 meters from the other bank of the river on their backward journey. What will be the width of the river (in meters)?

Options:
- **(A)** `1.3kms`
- **(B)** `400 m`
- **(C)** `1 km`
- **(D)** `850 m`

Correct Answer: (D) `850 m`

PrepInsta Detailed Explanation & Mathematical Derivation:

Let the two opposite ends of the river be X and Y and the distance

between them be D meters.(i.e., width = D meters)

Let P and Q be the two men starting from the opposite banks

(i.e., from X and Y respectively).

Let the speed of P and Q be A and B m/hr .

I meet :

During I meet, P travels 340m from X while Q travels

(D - 340)m from Y.

Therefore, Time taken for P to travel 340m = Time taken for

Q to travel (D - 340)

Or 340 / A = (D - 340) / B

Or 340 / (D - 340) = A / B ...(1)

II meet :

After crossing spot I, both of them proceed in their respective directions,

reach banks and return back to cross each other at

Spot II which is 170m from Y.

From Spot I to Spot II, P would had travelled a distance of

(D - 340) + 170 m

From Spot I to Spot II, Q would had travelled a distance of

340 + (D - 170) m

Time taken by P to travel from Spot I to Spot II will be the same

as that of Q from Spot I to Spot II

Therefore, A / (D - 340) + 170 = B / 340 + (D - 170)

Or

(D - 340) + 170 / 340 + (D - 170) = A / B ...(2)

From equations I and II, we get,

340 / (D - 340) = (D - 340) + 170 / 340 + (D - 170)

340 / (D - 340) =

D - 170 / D + 170

By Cross- Multiplying,

340 (D + 170) = (D - 170) (D - 340)

340D + 57800 =

D2 - 170D - 340D + 57800

D2 - 850D = 0

By Factorizing,

D(D - 850) = 0

D = 850

Hence the width of the river = 850 m

Question 10 (PrepInsta quiz-1 - Q10)

Question Stem:
When a train travels at a speed of 60kmph,it reaches the destination on time.when the same train travels at a speed of 50kmph, it reaches its destination 15min late.what is the length of journey?

Options:
- **(A)** `75km`
- **(B)** `50km`
- **(C)** `60km`
- **(D)** `85km`

Correct Answer: (A) `75km`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Let d be the length of the journey,


time taken to travel distance d at speed of 50 kmph = d/50


time taken to travel distance d at speed of 60 kmph = d/60


difference in time = 15 min = 15/60 hr


d/50 - d/60 = 15/60


d/5 - d/6 = 15/6


d/5 =15


d = 15*5


d = 75 km


 
```

Question 11 (PrepInsta quiz-1 - Q11)

Question Stem:
A girl goes to her office for work which is 50 miles. She goes to her office few distance by bicycle and remaining by train. The speed of bicycle is 15 mph and that of train is twice of the bicycle. If she spend 20 min. more on bicycle, then total time tak

Options:

- **(A)** `1 hr 30 min`
- **(B)** `2 hr 20 min`
- **(C)** `2 hr 30 min`
- **(D)** `2 hr 50 min`

Correct Answer: (B) `2 hr 20 min`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Let time travelled in train is x min then in cycle (x+20) min.


(30/60)x + (15/60)(x + 20) = 50 or x = 60


Total time taken 60 + (60 + 20) = 2 hr 20 min
```
